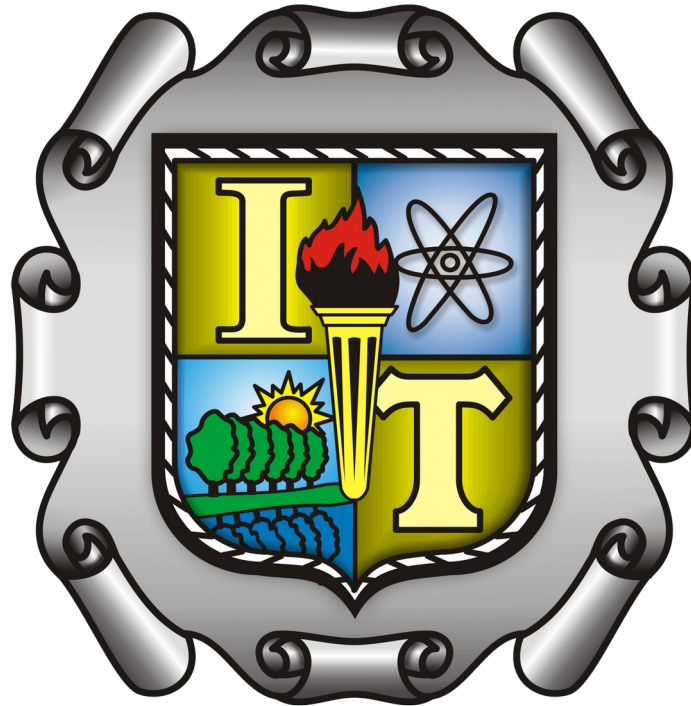


Tecnologico nacional de mexico

Instituto tecnologico de saltillo



5.8 Instalación del entorno LAMP en una máquina virtual en
infraestructura de la nube.

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Numero de control: 20051204

```

System information as of Thu Apr 30 03:35:12 UTC 2026

System load:  0.13           Processes:            150
Usage of /:   6.3% of 28.02GB Users logged in:      0
Memory usage: 39%           IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is enabled.

0 updates can be applied immediately.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@LAMP:~$
azureuser@LAMP:~$
azureuser@LAMP:~$
azureuser@LAMP:~$
azureuser@LAMP:~$
azureuser@LAMP:~$

```

descarga y actualizar

```

Processing triggers for systemd (253.4-1ubuntu8.15) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Setting up packagekit (1.2.8-2ubuntu1.5) ...
Setting up packagekit-tools (1.2.8-2ubuntu1.5) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager
systemctl restart chrony.service multipathd.service systemd-journald.service systemd-networkd.service systemd-resolv
d.service systemd-udev.service udisks2.service walinuxagent.service

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
azureuser @ session #2: apt[3987], sshd[3450]
azureuser @ user manager service: systemd[3455]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@LAMP:~$

```

descarga de apache

```
Enabling conf: serve-cgi-bin.
Enabling site: 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
azureuser @ session #2: sshd[3450]
azureuser @ user manager service: systemd[3455]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@LAMP:~$
```

apache corriendo

```
Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
azureuser @ session #2: sshd[3450]
azureuser @ user manager service: systemd[3455]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@LAMP:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-30 03:37:39 UTC; 38s ago
     Docs: https://httpd.apache.org/docs/2.4/
  Main PID: 10456 (apache2)
    Tasks: 55 (limit: 977)
   Memory: 5.3M (peak: 5.6M)
      CPU: 36ms
   CGroup: /system.slice/apache2.service
           └─10456 /usr/sbin/apache2 -k start
           └─10459 /usr/sbin/apache2 -k start
           └─10460 /usr/sbin/apache2 -k start

Apr 30 03:37:39 LAMP systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 30 03:37:39 LAMP systemd[1]: Started apache2.service - The Apache HTTP Server.
azureuser@LAMP:~$
```

descarga php

```
apache2_switch_mpm switch to mpm_event
apache2_invoke: Enable module php8.3
Setting up php8.3 (8.3.6-0ubuntu0.24.04.8) ...
Setting up libapache2-mod-php (2:8.3+93ubuntu2) ...
Setting up php (2:8.3+93ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for php8.3-cli (8.3.6-0ubuntu0.24.04.8) ...
Processing triggers for libapache2-mod-php8.3 (8.3.6-0ubuntu0.24.04.8) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
azureuser @ session #2: sshd[3450]
azureuser @ user manager service: systemd[3455]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@LAMP:~$
```

PHP corriendo

PHP Version 8.3.6



System	Linux LAMP 6.17.0-1011-azure #11-24.04.2-Ubuntu SMP Wed Mar 25 22:46:36 UTC 2026 x86_64
Build Date	Mar 20 2026 02:32:55
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.3/apache2
Loaded Configuration File	/etc/php/8.3/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.3/apache2/conf.d
Additional .ini files parsed	/etc/php/8.3/apache2/conf.d/10-mysqlnd.ini, /etc/php/8.3/apache2/conf.d/10-opcache.ini, /etc/php/8.3/apache2/conf.d/10-pdo.ini, /etc/php/8.3/apache2/conf.d/20-calendar.ini, /etc/php/8.3/apache2/conf.d/20-ctype.ini, /etc/php/8.3/apache2/conf.d/20-exif.ini, /etc/php/8.3/apache2/conf.d/20-ffi.ini, /etc/php/8.3/apache2/conf.d/20-fileinfo.ini, /etc/php/8.3/apache2/conf.d/20-ftp.ini, /etc/php/8.3/apache2/conf.d/20-gettext.ini, /etc/php/8.3/apache2/conf.d/20-iconv.ini, /etc/php/8.3/apache2/conf.d/20-mysqli.ini, /etc/php/8.3/apache2/conf.d/20-pdo_mysqli.ini, /etc/php/8.3/apache2/conf.d/20-phar.ini, /etc/php/8.3/apache2/conf.d/20-posix.ini, /etc/php/8.3/apache2/conf.d/20-readline.ini, /etc/php/8.3/apache2/conf.d/20-shmop.ini, /etc/php/8.3/apache2/conf.d/20-sockets.ini, /etc/php/8.3/apache2/conf.d/20-sysvmsg.ini, /etc/php/8.3/

instala mySQL

```
reading /usr/share/mecab/dic/ipadic/Noun.csv ... 146
reading /usr/share/mecab/dic/ipadic/Postp.csv ... 146
reading /usr/share/mecab/dic/ipadic/Conjunction.csv ... 171
reading /usr/share/mecab/dic/ipadic/Postp-col.csv ... 91
reading /usr/share/mecab/dic/ipadic/Noun.name.csv ... 34202
reading /usr/share/mecab/dic/ipadic/Symbol.csv ... 208
reading /usr/share/mecab/dic/ipadic/Prefix.csv ... 221
reading /usr/share/mecab/dic/ipadic/Noun.number.csv ... 42
reading /usr/share/mecab/dic/ipadic/Interjection.csv ... 252
reading /usr/share/mecab/dic/ipadic/Noun.demonst.csv ... 120
reading /usr/share/mecab/dic/ipadic/Others.csv ... 2
reading /usr/share/mecab/dic/ipadic/Noun.nai.csv ... 42
reading /usr/share/mecab/dic/ipadic/Noun.verbal.csv ... 12146
reading /usr/share/mecab/dic/ipadic/Verb.csv ... 130750
reading /usr/share/mecab/dic/ipadic/Noun.proper.csv ... 27328
reading /usr/share/mecab/dic/ipadic/Noun.adverbial.csv ... 795
reading /usr/share/mecab/dic/ipadic/Adj.csv ... 27210
reading /usr/share/mecab/dic/ipadic/Noun.place.csv ... 72999
reading /usr/share/mecab/dic/ipadic/Adnominal.csv ... 135
reading /usr/share/mecab/dic/ipadic/Noun.adjv.csv ... 3328
reading /usr/share/mecab/dic/ipadic/Adverb.csv ... 3032
reading /usr/share/mecab/dic/ipadic/Suffix.csv ... 1393
reading /usr/share/mecab/dic/ipadic/Filler.csv ... 19
reading /usr/share/mecab/dic/ipadic/Auxil.csv ... 199
reading /usr/share/mecab/dic/ipadic/Noun.others.csv ... 151
emitting double-array: 6% |##
Progress: [ 86%] [#####.....]
```

Acceder a MySQL

```
All done!
azureuser@LAMP:~$ sudo mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.45-0ubuntu0.24.04.1 (Ubuntu)

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> █
```

crear una tabla

```
mysql> CREATE DATABASE prueba_lamp;
Query OK, 1 row affected (0.02 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| prueba_lamp |
| sys |
+-----+
5 rows in set (0.01 sec)

mysql> EXIT; █
```

Crear un Nano

```
bye  
azureuser@LAMP:~$ sudo nano /var/www/html/testdb.php  
azureuser@LAMP:~$
```

Conectado a Mysql

Conectado exitosamente a MySQL 🎉